08/10/25

UNIT-2

RELATIONS & FUNCTIONS

• CARTESIAN PRODUCT OF THE SETS:

→ Let A & B, be any 2 sets, then the set of all ordered pairs (a, b), where $a \in A$, $b \in B$ is called the cartesian product or cross-product of A & B.

→ It is denoted by $A \times B$.

→ i.e., $A \times B = \{(a, b) \mid a \in A, b \in B\}$

→ Also, $B \times A = \{(b, a) \mid b \in B, a \in A\}$

★ NOTE: ⇒ (i) $A \times B \neq B \times A$.

(ii) If $B = A$, then $A \times A = A^2 = \{(a, a) \mid a \in A\}$

(iii) If A & B are finite sets, then order of A is given by - $|A| = m$ & order of B is - $|B| = n$, then order of $A \times B$ - $|A \times B| = m \times n$.

eg-

Q1) Let $A = \{1, 3, 5\}$

$B = \{2, 3\}$

$C = \{4, 6\}$

Find the following - (i) $A \times B$

(ii) $B \times A$

(iii) $(A \cup B) \times C$

(iv) $(A \times B) \cup C$

(v) $(A \times B) \cap (B \times A)$

(vi) $A \times C$

Solⁿ - (i) $A \times B = \{(1,2), (1,3), (3,2), (3,3), (5,2), (5,3)\}$

(ii) $B \times A = \{(2,1), (2,3), (2,5), (3,1), (3,3), (3,5)\}$

(iii) $A \cup B = \{1, 2, 3, 5\}$

$(A \cup B) \times C = \{(1,4), (1,6), (2,4), (2,6), (3,4), (3,6), (5,4), (5,6)\}$

(iv) $(A\times B)\cup C$

$A\times B=\{(1,2),(1,3),(3,2),(3,3),(5,2),(5,3)\}$

$(A\times B)\cup C=\{(1,2),(1,3),(3,2),(3,3),(5,2),(5,3),4,6\}$

(v) $(A\times B)\cap(B\times A)=\{(3,3)\}$

(vi) $A\times C=\{(1,4),(1,6),(3,4),(3,6),(5,4),(5,6)\}$

Q2) Let $A=\{a,b\}$
$B=\{1,2,3\}$

Find (i) $A\times B$
(ii) $B\times A$
(iii) $A\times A$
(iv) $B\times B$

Solⁿ - (i) $A\times B=\{(a,1),(a,2),(a,3),(b,1),(b,2),(b,3)\}$

(ii) $B\times A=\{(1,a),(1,b),(2,a),(2,b),(3,a),(3,b)\}$

(iii) $A\times A=\{(a,a),(a,b),(b,a),(b,b)\}$

(iv) $B\times B=\{(1,1),(1,2),(1,3),(2,1),(2,2),(2,3),(3,1),(3,2),(3,3)\}$.

Q3) For non-empty sets A, B, C, prove that -

(i) $A\times(B\cup C)=(A\times B)\cup(A\times C)$

HW (ii) $(A\cup B)\times C=(A\times C)\cup(B\times C)$

(iii) $A\times(B\cap C)=(A\times B)\cap(A\times C)$

HW (iv) $(A\cap B)\times C=(A\times C)\cap(B\times C)$

(v) $A\times(B-C)=(A\times B)-(A\times C)$

Solⁿ - (i) Let $(x,y)\in\{A\times(B\cup C)\}\Leftrightarrow\{x\in A \text{ \& } y\in B\cup C\}$

$\Leftrightarrow\{x\in A \text{ and } (y\in B \text{ or } y\in C)\}$

$\Leftrightarrow\{(x\in A \text{ and } y\in B) \text{ or } (x\in A \text{ and } y\in C)\}$

$\Leftrightarrow\{(x,y)\in A\times B \text{ or } (x,y)\in A\times C\}$

$\Leftrightarrow (x,y)\in(A\times B)\cup(A\times C)$

i.e. $A\times(B\cup C)=(A\times B)\cup(A\times C)$ //

(iii) $A\times(B\cap C)=(A\times B)\cap(A\times C)$

Let $\{(x,y)\in\{A\times(B\cap C)\}\Leftrightarrow\{x\in A \text{ or } y\in B\cap C\}$

$\Leftrightarrow\{x\in A \text{ or } (y\in B \text{ and } y\in C)\}$

$\Leftrightarrow\{(x\in A \text{ or } y\in B) \text{ and } ((x\in A) \text{ or } y\in C)\}$

$\Leftrightarrow\{(x,y)\in A\times B \text{ and } (x,y)\in A\times C\}$

$\Leftrightarrow (x,y)\in(A\times B)\cap(A\times C)$ //

i.e. $A\times(B\cap C)=(A\times B)\cap(A\times C)$ /

(ii) $(A\cup B)\times C=(A\times C)\cup(B\times C)$

Let $(x,y)\in\{(A\cup B)\times C\}\Leftrightarrow\{x\in(A\cup B) \text{ \& } y\in C\}$

$\Leftrightarrow\{(x\in A \text{ or } x\in B) \text{ and } y\in C\}$

$\Leftrightarrow\{(x\in A \text{ and } y\in C) \text{ or } ((x\in B) \text{ and } y\in C)\}$

$\Leftrightarrow\{(x,y)\in A\times C \text{ or } (x,y)\in B\times C\}$

$\Leftrightarrow (x,y)\in(A\times C)\cup(B\times C)$

i.e. $(A\cup B)\times C=(A\times C)\cup(B\times C)$ //

(iv) $(A\cap B)\times C = (A\times C)\cap(B\times C)$

Let $(x,y)\in\{(A\cap B)\times C\} \Leftrightarrow \{x\in A\cap B \text{ or } y\in C\}$

$\Leftrightarrow \{(x\in A \text{ and } x\in B) \text{ or } y\in C\}$

$\Leftrightarrow \{(x\in A \text{ or } y\in C) \text{ and } ((x\in B) \text{ or } y\in C)\}$

$\Leftrightarrow \{(x,y)\in A\times C \text{ and } (x,y)\in B\times C\}$

$\Leftrightarrow (x,y)\in (A\times C)\cap(B\times C)$

i.e. $(A\cap B)\times C = (A\times C)\cap(B\times C)$ //

$\boxed{A-B = A\cap\overline{B}}$

(v) $A\times(B-C) = (A\times B)-(A\times C)$

Let $(x,y)\in\{A\times(B-C)\} \Leftrightarrow \{x\in A \text{ and } y\in(B-C)\}$

$\Leftrightarrow \{x\in A \text{ and } y\in(B\cap\overline{C})\}$

$\Leftrightarrow \{x\in A \text{ and } (y\in B \text{ and } y\notin C)\}$

$\Leftrightarrow \{(x\in A \text{ and } y\in B) \text{ and } (x\in A) \text{ and } y\notin C)\}$

$\Leftrightarrow \{(x,y)\in A\times B \text{ and } (x,y)\notin A\times C\}$

$\Leftrightarrow \{(x,y)\in A\times B\cap(\overline{A\times C})\}$

i.e. $A\times(B-C) = (A\times B)-(A\times C)$.

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Q) Let $A=\{1,2,3\}$, $B=\{2,3,4\}$, $C=\{3,4,5\}$. Verify that-

(i) $A\times(B\cup C) = (A\times B)\cup(A\times C)$

(ii) $A\times(B\cap C) = (A\times B)\cap(A\times C)$

Solⁿ- By data,

$A=\{1,2,3\}$ $B=\{2,3,4\}$ $C=\{3,4,5\}$

(i) We have, $B\cup C = \{2,3,4,5\}$

i.e. $A\times(B\cup C) = \{(1,2),(1,3),(1,4),(1,5),(2,2),(2,3),(2,4),(2,5),(3,2),(3,3),(3,4),(3,5)\} \rightarrow (1)$

Also, $A\times B = \{(1,2),(1,3),(1,4),(2,2),(2,3),(2,4),(3,2),(3,3),(3,4)\}$

$A\times C = \{(1,3),(1,4),(1,5),(2,3),(2,4),(2,5),(3,3),(3,4),(3,5)\}$

$(A\times B)\cup(A\times C) = \{(1,2),(1,3),(1,4),(1,5),(2,2),(2,3),(2,4),(2,5),(3,2),(3,3),(3,4),(3,5)\} \rightarrow (2)$

(1) = (2), $\therefore A\times(B\cup C) = (A\times B)\cup(A\times C)$ //

(ii) We have, $B\cap C = \{3,4\}$

i.e., $A\times(B\cap C) = \cancel{\{(1,4),(2,4),(3,4)\}}\ \{(1,3),(2,3),(3,3),(1,4),(2,4),(3,4)\}$

$A\times B = \{(1,2),(1,3),(1,4),(2,2),(2,3),(2,4),(3,2),(3,3),(3,4)\}$ - (1)

$A\times C = \{(1,3),(1,4),(1,5),(2,3),(2,4),(2,5),(3,3),(3,4),(3,5)\}$

$(A\times B)\cap(A\times C) = \{(1,3),(1,4),(2,3),(2,4),(3,3),(3,4)\}$ // - (2)

(1) = (2), $\therefore A\times(B\cap C) = (A\times B)\cap(A\times C)$ //

Hw

Q) If $A=\{1,2,3,4\}$ $B=\{3,4,5,6\}$ $C=\{2,4,6\}$. Verify that,-

(i) $(A\cup B)\times C=(A\times C)\cup(B\times C)$

(ii) $(A\cap B)\times C=(A\times C)\cap(B\times C)$

*RELATIONS:

→ Let A & B be 2 sets. Then, the subset of $A\times B$ is called relation from A to B.

→ i.e. $R\subseteq A\times B$

where, $A\times B=\{(a,b) \mid a\in A, b\in B)\}$

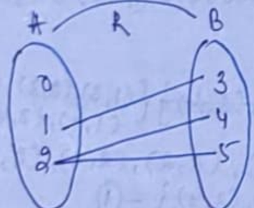
→ i.e., if $(a,b)\in R$, then we say that 'a' is related to 'b' by R and denoted by aRb.

→ If R is a subset of $A\times A$ ($R\subseteq A\times A$), then R is called binary relation on A.

eg- If $A=\{0,1,2\}$ $B=\{3,4,5\}$ & $R=\{(1,3),(2,4),(2,5)\}$

→ We note that, $R\subseteq A\times B$ i.e., R is a relation from A to B and 1R3, 2R4, 2R5.

→ This relation can be represented in a diagram as shown below-



→ Domain of $R=\{x\in A \mid xRy \text{ for some } x\in A\}$

→ Codomain (or) Range of $R=\{y\in B \mid xRy \text{ for some } y\in B\}$

• Matrix of a Relation

→ Let $A=\{a_1, a_2, a_3 \ldots\ldots a_m\}$
$B=\{b_1, b_2, b_3 \ldots\ldots b_n\}$
be any 2 finite sets.

→ Let R be the relation from A to B. Then, the matrix of the relation R is defined as-

$$M(R)=[m_{ij}]_{m\times n}$$ that is an $m\times n$ matrix

where, $m_{ij}=\begin{cases}1, & \text{if } a_i R b_j\\ 0, & \text{if } a_i \not R b_j\end{cases}$

→ The rows of M(R) corresponds to the order of A and columns to the order of B.

*NOTE ⇒ If 'A' is a set with 'm' elements & B is a set with 'n' elements, then the no. of relations from A to B is given by - 2^{mn}.

Q1) Let $A=\{1,2,3,4,5\}$. Let R be the relation on A defined as - aRb if $a<b$. List all the elements of R and write its domain & range.

Sol^n- Given, $A=\{1,2,3,4,5\}$

$$A\times A=\{(1,1),(1,2),(1,3),(1,4),(1,5),(2,1),(2,2),(2,3),(2,4),(2,5),(3,1),(3,2),(3,3),(3,4),(3,5),(4,1),(4,2),(4,3),(4,4),(4,5),(5,1),(5,2),(5,3),(5,4),(5,5)\}$$

$R=\{(a,b) \mid aRb \text{ if } a<b\}$

$R=\{(1,2),(1,3),(1,4),(1,5),(2,3),(2,4),(2,5),(3,4),(3,5),(4,5)\}$

Domain - $R=\{1,2,3,4\}$

Range - $R=\{2,3,4,5\}$

$$M[R] = \begin{array}{c|ccccc} & 1 & 2 & 3 & 4 & 5 \\ \hline 1 & & & & & \\ 2 & & & & & \\ 3 & & & & & \\ 4 & & & & & \\ 5 & & & & & \end{array} \begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Q2) Let $A=\{1,2,3\}$, $B=\{2,4,5\}$. Determine-
(i) Order of $A \times B$ / $|A \times B|$
(ii) No. of relations from A to B.
(iii) No. of binary relations on A.

Solⁿ- $A=\{1,2,3\}$, $|A|=3$
$B=\{2,4,5\}$, $|B|=3$

(i) $|A \times B| = mn = 3 \times 3 = 9$
(ii) No. of relations from A to B - $2^{mn} = 2^{3\times 3} = 2^9 = 512$
(iii) No. of binary relations from A to A - $2^{mm} = 2^9 = 512$

Q3) Let A & B be the finite sets with order of $B=3$ ($|B|=3$). If there are 4096 relations from A to B. What is the order of A?

Solⁿ- $|B|=3$
$\hookrightarrow n=3$

$2^{mn} = 4096$
$2^{3m} = 4096$
$2^{3m} = 2^{12}$

Equating powers with same base,
$3m = 12$
$\boxed{m=4}$

$\therefore |A| = 4$

Q4) Let $A=\{1,2,3,4,6\}$ and R be the relation defined on A by aRb iff a is a multiple of 'b'. Represent the relation R as a matrix.

Solⁿ- Given, $A=\{1,2,3,4,6\}$

$A \times A = \{(1,1),(1,2),(1,3),(1,4),(1,6),(2,1),(2,2),(2,3),(2,4),(2,6),(3,1),(3,2),(3,3),(3,4),(3,6),(4,1),(4,2),(4,3),(4,4),(4,6),(6,1),(6,2),(6,3),(6,4),(6,6)\}$

$R=\{(a,b) \mid a$ is a multiple of $b\}$
$R = \{(1,1),(2,1),(2,2),(3,1),(3,3),\cancel{(3,6)}$ $(4,1),(4,4),(6,1),(6,6),(6,2),(6,3),(4,2)\}$

$$M[R] = \begin{array}{c|ccccc} & 1 & 2 & 3 & 4 & 6 \\ \hline 1 & & & & & \\ 2 & & & & & \\ 3 & & & & & \\ 4 & & & & & \\ 6 & & & & & \end{array} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 \end{bmatrix}$$

Q5) If $A=\{1,2,3,4\}$ & $B=\{1,4,6,9\}$. Let R be the relation defined by ~~ARB~~ aRb iff $b=a^2$. Find the relation R, domain, range and matrix of the relation.

Solⁿ- $A \times B = \{(1,1),(1,4),(1,6),(1,9),(2,1),(2,4),(2,6),(2,9),(3,1),(3,4),(3,6),(3,9),(4,1),(4,4),(4,6),(4,9)\}$

$R = \{(1,1),(2,4),(3,9)\}$

Domain - $R \to \{1,2,3\}$
Range - $R \to \{1,4,9\}$

Q6) Let $A=\{1,2,3,4\}$ and R_1, R_2, R_3 are the relations of A defined as follows -

(i) aR_1b iff $a\le b$

(ii) aR_2b iff $a>b$

(iii) aR_3b iff a is odd & b is even.

Write down these relations as a set of ordered pairs and the corresponding matrix $M[R]$.

Solⁿ - $A=\{1,2,3,4\}$

$R_1=\{(1,1),(1,2),(1,3),(1,4),(2,2),(2,3),(2,4),(3,3),(3,4),(4,4)\}$

$R_2=\{(2,1),(3,1),(3,2),(4,1),(4,2),(4,3)\}$

$R_3=\{(1,2),(1,4),(3,2),(3,4)\}$

$$M[R_1]=\begin{array}{c c} & \begin{array}{cccc}1&2&3&4\end{array}\\ \begin{array}{c}1\\2\\3\\4\end{array} & \begin{bmatrix}1&1&1&1\\0&1&1&1\\0&0&1&1\\0&0&0&1\end{bmatrix}\end{array}$$

$$M[R_2]=\begin{array}{c c} & \begin{array}{cccc}1&2&3&4\end{array}\\ \begin{array}{c}1\\2\\3\\4\end{array} & \begin{bmatrix}0&0&0&0\\1&0&0&0\\1&1&0&0\\1&1&1&0\end{bmatrix}\end{array}$$

$$M[R_3]=\begin{array}{c c} & \begin{array}{cccc}1&2&3&4\end{array}\\ \begin{array}{c}1\\2\\3\\4\end{array} & \begin{bmatrix}0&1&0&1\\0&0&0&0\\0&1&0&1\\0&0&0&0\end{bmatrix}\end{array}$$

→ Let A be a finite set and R be the binary relation on A and R can be represented pictorially as follows-

(i) Draw a small circle for each element of A and label the circle with the corresponding elements of A. These circles are called vertices.

(ii) Draw an arrow from a_i to a_j if $a_i R a_j$, this arrow is called edge.

(iii) The resulting picture represents the relation R called as directed graph or digraph.

(iv) In a digraph,

→ A vertex from which an edge leaves is called origin or source of that edge.

→ A vertex where an edge end is called terminus for that edge.

→ A vertex which is neither a source nor a termi[nus] is called isolated vertex.

→ An edge for which the source and the terminus are same is called loop.

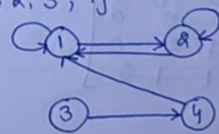
→ In-degree of a vertex: No. of edges terminating at vertex.

→ out-degree of a vertex: No. of edges leaving a vertex

Q1) Let $A=\{1,2,3,4\}$ and $R=\{(1,1),(1,2),(2,1),(2,2),(3,4),(4,1)\}$ be a relation of A. Write down the digraph and matrix of the relation. Also determine the in and out degrees of the vertices.

Solⁿ - $A=\{1,2,3,4\}$

Digraph of R:



Matrix of R = $\begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array}\begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \end{array}$

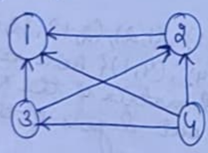
Elements	In-Degree	Out degree
1	3	2
2	2	2
3	0	1
4	1	1

Q6)

Q) If R ~~equal~~ = $\{(x, y) / x > y\}$ is a relation defined on the set $A = \{1, 2, 3, 4\}$.
(i) Write down R as a set of ordered pairs.
(ii) Write down the digraph and matrix of R.
(iii) Determine the in & out degree of the vertices in the digraph.

Sol^n - $A = \{1, 2, 3, 4\}$
(i) $R = \{(2,1), (3,1), (3,2), (4,1), (4,2), (4,3)\}$

(ii) Digraph of R:



Matrix of R = $\begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array}\begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix} \end{array}$

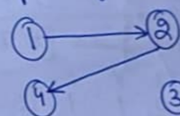
(iii)

Elements	In-degree	Out-degree
1	3	0
2	2	1
3	1	2
4	0	3

Q) $A = \{1, 2, 3, 4\}$ and R be the relation of A defined by xRy iff $y = 2x$.
(i) Write down R as a set of ordered pairs.
(ii) Write down the digraph and matrix of R.
(iii) Determine the in & out degree of the vertices in the digraph.

Sol^n - (i) $R = \{(1, 2), (2, 4)\}$

(ii) Digraph of R:



3 → Isolated vertex

Matrix of R: $\begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array}\begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$

(iii)

Elements	In degree	Out degree
1	0	1
2	1	1
3	0	0
4	1	0

Q) A = {1, 2, 3, 4} and let R be the
by xRy iff x divides y. (y|x)

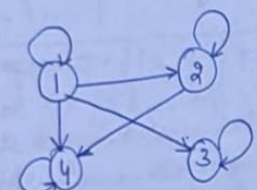
(i) Write R

(ii) Digraph & matrix of R

(iii) In and out degree.

Solⁿ

(i) R = ~~{(1,1), (2,1), (2,2),~~ {(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)}

(ii) 

Digraph

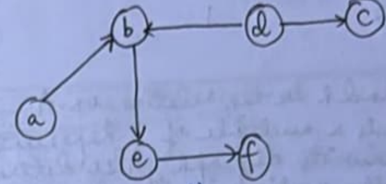
Matrix-

$$M[R] = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{array}$$

(iii)

Element	In-degree	Out-degree
1	1	4
2	2	2
3	2	1
4	3	1

Q) For set ~~A, B, C, D upto~~ A = {a, b, c, d, e, f}, the digraph given below represents a relation R on A. Determine R as an associated relation and write the matrix of R.



A = {a, b, c, d, e, f}

Solⁿ- R = {(a, b), (b, e), (d, b), (d, c), (e, f)}

$$M[R] = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \\ f \end{array} \begin{array}{c} \begin{array}{cccccc} a & b & c & d & e & f \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

Q) Let A = {2, 4, 5, 7} and let R be the relation on A having the matrix -

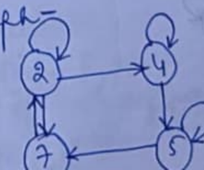
$$M_R = \begin{array}{c} \\ 2 \\ 4 \\ 5 \\ 7 \end{array} \begin{array}{c} \begin{array}{cccc} 2 & 4 & 5 & 7 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

(i) Write the relation R and draw its digraph.

(ii) Determine the in & out degree of all the vertices.

Solⁿ- (i) R = {(2, 2), (2, 4), (2, 7), (4, 4), (4, 5), (5, 5), (5, 7), (7, 2)}

Digraph-



(ii)

Element	In-degree	Out-degree
2	2	3
4	2	2
5	2	2
7	2	1

Q) Let $A=\{1,2,3,4,6\}$ and R be the relation on A defined by aRb iff a is a multiple of b. Represent R as a matrix and draw its digraph. Also determine the in & out degree of the vertices in the digraph

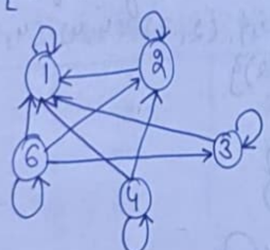
Soln- $R=\{(1,1),(2,1),(2,2),(3,1),(3,3),(4,1),(4,2),(4,4),(6,1),(6,2),(6,3),(6,6)\}$

~~Matrix:~~

Matrix: M[R]

$$\begin{array}{c|c} & \begin{matrix}1 & 2 & 3 & 4 & 6\end{matrix} \\ \hline \begin{matrix}1\\2\\3\\4\\6\end{matrix} & \begin{bmatrix}1 & 0 & 0 & 0 & 0\\1 & 1 & 0 & 0 & 0\\1 & 0 & 1 & 0 & 0\\1 & 1 & 0 & 1 & 0\\1 & 1 & 1 & 0 & 1\end{bmatrix}\end{array}$$

Digraph:



Element	In-degree	Out-degree
1	5	1
2	3	2
3	2	~~2~~ 3
4	1	~~2~~ 4
6	~~0~~ 1	4

* PROPERTIES OF RELATION:

1. REFLEXIVE:
→ A relation 'R' on a set A is said to be reflexive if $\forall a \in A, (a,a) \in R$.
→ A relation 'R' on a set A is said to be irreflexive if $\forall a \in A, (a,a) \notin R$.

2. SYMMETRIC:
→ A relation 'R' on a set A is said to be symmetric if $(a,b) \in R$ then $(b,a) \in R$, $\forall a,b \in A$.
→ A relation 'R' on a set A is said to be not symmetric if $(a,b) \in R$ then $(b,a) \notin R$, $\forall a,b \in A$.

3. ANTI-SYMMETRIC:
→ A relation 'R' on a set A is said to be anti-symmetric if $(a,b) \in R$ and $(b,a) \in R$ then $a=b$, $\forall a,b \in A$ (or)
if $(a,b) \in R$ & $(b,a) \notin R$ then $a \neq b$, $\forall a,b \in A$.

4. TRANSITIVE: R
→ A relation 'R' on a set 'A' is said to be transitive if $(a,b) \in R$ and $(b,c) \in R$ then $(a,c) \in R$, $\forall a,b,c \in A$.

5. EQUIVALENCE:
→ A relation 'R' on a set A is said to be equivalence relation if R is reflexive, symmetric & transitive.

6. PARTIAL ORDER:
→ A relation 'R' on a set A is said to be partial order relation if R is reflexive, antisymmetric & transitive.

Q) Let $A=\{1,2,3\}$. Determine the nature of the following relations on A.

(i) $R_1=\{(1,2),(2,1),(1,3),(3,1)\}$

R_1 is symmetric

(ii) $R_2=\{(1,1),(2,2),(3,3),(2,3)\}$

R_2 is reflexive & anti-symmetric

(iii) $R_3=\{(1,1),(2,2),(3,3),(2,3),(3,2)\}$

R_3 is reflexive & symmetric

(iv) $R_4=\{(2,3),(3,4),(2,4)\}$

R_4 is transitive & anti-symmetric.

Q) Let $A=\{1,2,3\}$

$R=\{(1,1),(2,2),(1,2),(2,3),(3,3)\}$ be a relation on A. Check whether R is equivalence or not.

Solⁿ- (i) Reflexive -

$\forall a\in A,\ (a,a)\in R$.

We have $(1,1),(2,2),(3,3)\in R$.

$\therefore$ R is reflexive.

(ii) Symmetric -

$\forall$ a, b $\in A$, if $(a,b)\in R$ then $(b,a)\in R$

$(1,2)\in R$ but $(2,1)\notin R$

$(2,3)\in R$ but $(3,2)\notin R$.

$\therefore$ R is not symmetric.

(iii) Transitive -

if $(a,b)\in R$ & $(b,c)\in R$ then $(a,c)\in R \Rightarrow (a,c)\in R$

$(1,2)\in R$ & $(2,3)\in R \Rightarrow (1,3)\notin R$.

$\therefore$ R is not transitive.

Hence, R is not equivalence.

Q) Let $A=\{1,2,3\}$

$R=\{(1,1),(1,2),(1,3),(2,1),(2,2),(2,3),(3,1),(3,2),(3,3)\}$ be a relation on A.

Verify that R is a equivalence relation.

Solⁿ- (i) Reflexive -

$\forall a\in A,\ (a,a)\in R$

We have $(1,1),(2,2),(3,3)\in R$.

$\therefore$ R is reflexive.

(ii) Symmetric -

$\forall a,b\in A$, if $(a,b)\in R$ then $(b,a)\in R$

$(1,2)\in R$ then $(2,1)\in R$, $(1,3)\in R$ then $(3,1)\in R$

$(2,3)\in R$ then $(3,2)\in R$.

$\therefore$ R is symmetric.

(iii) Transitive -

if $(a,b)\in R$ & $(b,c)\in R$ then $(a,c)\in R \Rightarrow (a,c)\in R$.

$(1,2)\in R$ & $(2,3)\in R \Rightarrow (1,3)\in R$.

$(1,3)\in R$ & $(3,2)\in R \Rightarrow (1,2)\in R$

$\therefore$ R is transitive.

Hence, R is equivalence.

Q) Let $A=\{1,2,3,4\}$ and $R=\{(1,1),(1,2),(2,1),(2,2),(3,1),(3,3),(1,3),(4,1),(4,4)\}$ be a relation on A.

Is R equivalence?

Solⁿ: Given, $A=\{1,2,3,4\}$

$R=\{(1,1),(1,2),(2,1),(2,2),(3,1),(3,3),(1,3),(4,1),(4,4)\}$

(i) Reflexive -

$\forall a\in A,\ (a,a)\in R$

We have $(1,1),(2,2),(3,3),(4,4)\in R$.

$\therefore$ R is reflexive.

(ii) Symmetric -

$\forall a,b\in A$, if $(a,b)\in R$ then $(b,a)\in R$

$(1,2)\in R$, $(2,1)\in R$ $(4,1)\in R$, $(1,4)\notin R$.

$(2,1)\in R$, $(1,3)\in R$

∴ R is not symmetric.

(ii) Hence, it is not equivalence.

Q) Let $A = \{1, 2, 3, 4\}$ and R_1 & R_2 be the relations on A as given below-

$R_1 = \{(1, 1), (2, 1), (2, 2), (3, 3), (4, 4), (4, 3)\}$

$R_2 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 1), (3, 3), (1, 4), (4, 1), (4, 4)\}$

Check whether R_1 & R_2 are equivalence relation or not.

Sol^n^- R_1:

(i) Reflexive-

$\forall a \in A, (a, a) \in R$

We have $(1, 1), (2, 2), (3, 3) \in R$

∴ R_1 is reflexive

(ii) Symmetric-

$\forall a, b \in A$, if $(a, b) \in R$ then $(b, a) \in R$

$(2, 1) \in R$ but $(1, 2) \notin R$

$(4, 3) \in R$ but $(3, 4) \notin R$.

∴ R_1 is not symmetric

Hence, R_1 is not equivalence.

R_2:

(i) Reflexive-

$1R1$, $2R2$, $3R3$, $4R4$.

∴ R_2 is reflexive.

(ii) Symmetric-

$1R2$ & $2R1$

$3R1$ but $1 \not R 3$

$1R4$ & $4R1$

∴ R_2 is not symmetric.

Hence, R_2 is not equivalence.

Q) Let $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ and R be the relation defined on A by $(x, y) \in R$ iff $x - y$ is a multiple of 5. Verify that R is an equivalence relation.

Sol^n^- Given, $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

$R = \{(x, y) \mid x - y \text{ is a multiple of } 5\}$

(i) Reflexive:

For any, $x \in A$, we have-

$x - x = 0$

0 is a multiple of 5.

∴ $(x, x) \in R$ and R is reflexive

(ii) Symmetric:

For any ~~$(x,y) \in A$~~ $x, y \in A$, we have $(x, y) \in R$

So that $x - y = 5k$, where k is some integer.

Then, $y - x = -5k = -k(5)$, for some k.

∴ $(y, x) \in R$

∴ R is symmetric.

(iii) Transitive:

For any $x, y, z \in A$ & if $(x, y) \in R$ & $(y, z) \in R$.

Then, $x - y = 5k_1$ $y - z = 5k_2$, where k_1 & k_2 are integers.

Now, $x - z = x - y + y - z = 5k_1 + 5k_2 = 5(k_1 + k_2)$

$= 5k'$, where k' is $k_1 + k_2$

$\Rightarrow (x, z) \in R$.

∴ R is transitive.

Hence, R is an equivalence relation.

* PARTIAL ORDER RELATION & POSET:

→ A relation R on a set A is said to be partial order relation if R is reflexive, anti-symmetric and transitive.

→ A set A with partial order relation R defined on A is called partially ordered set or POSET.

→ It is denoted by a pair (A, R).

* HASSE DIAGRAM:

→ The graph/diagram of a partial order relation defined on a finite set is called Hasse diagram

(or)

Digraph of partial order relation.

→ To obtain Hasse diagram, we follow certain rules -

① Start with a directed graph of the relation placing the vertices on the page so that all the arrows point upward.

② Eliminate -

(i) The loops at all the vertices.

(ii) All the arrows whose existence is implied by transitive property.

(iii) The direction indicator on the arrows.

③ The digraph of the partial order drawn by adopting the rules is called POSET diagram | HASSE diagram.

Q) $A = \{1, 2, 3, 4\}$

$R = \{(1,1), (1,2), (2,2), (2,4), (1,3), (3,3), (3,4), (1,4), (4,4)\}$

Verify that R is a partial order relation ~~of~~ on A and write down its Hasse diagram.

Solⁿ - Reflexive -

1R1, 2R2, 3R3, 4R4

∴ R is reflexive.

Anti-Symmetric -

1R2, 2R4, 1R3, 3R4, 1R4 with $a \neq b$.

∴ R is anti-symmetric.

Transitive -

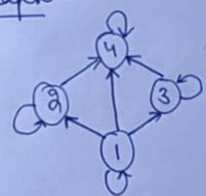
1R2 & 2R4 ⇒ 1R4

1R3 & 3R4 ⇒ 1R4

∴ R is transitive

Hence, (A, R) is a POSET | Partial order relation.

Digraph -



Hasse diagram -

→ Place the nos in the upward order

→ remove loops

→ remove directions

→ transitive property should be removed